EGC 301P / Operating Systems Lab Mini Project Guidelines

1. Objective

The objective of this mini project is to provide hands-on experience with core **Operating Systems concepts** through the design and implementation of a practical system/application. Students are expected to build a project that demonstrates real-world usage of OS principles such as concurrency, communication, and synchronization.

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2. Project Format

- This is an **individual project**.
- Each student must independently design and implement their own system/application.
- Any form of code sharing or collaboration in implementation is **not allowed**.

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3. Project Topic Selection

- Students are free to **choose their own project topics**.
- However, the chosen project **must incorporate all the mandatory OS concepts** listed below.
- Students are **strongly encouraged to choose unique and innovative project ideas**.

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4. Mandatory Concepts to be Covered

Every project must demonstrate the following concepts:

4.1 Role-Based Authorization

- Implement user roles (e.g., admin, user, guest).
- Define access control based on roles.
- Restrict operations depending on permissions.

4.2 File Locking

- Ensure safe file access when multiple processes/threads interact with shared files.
- Implement mechanisms such as:
  - Advisory or mandatory locking
  - Read/write locks

4.3 Concurrency Control

- Handle multiple processes/threads executing simultaneously.
- Use synchronization techniques such as:
  - Mutexes
  - Semaphores

4.4 Data Consistency

- Maintain correctness of shared data under concurrent access.
- Prevent issues like:
  - Race conditions

- Dirty reads
- Lost updates

4.5 Socket Programming

- Implement communication between systems/processes using sockets.
- Should include a client-server model

4.6 Inter-Process Communication (IPC)

- Demonstrate communication between processes using at least one IPC mechanism:
  - Pipes (named/unnamed)
  - Shared memory
  - Message queues
  - Signals

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5. Registration Process

- An **Excel sheet** will be released for project registration.
- Each student must fill in:
  - Name and ID
  - Selected project topic
  - Brief project description
- Changes in the sheet will be tracked and sabotaging the registration process will result in negative marks.

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6. Project Requirements

Each project must include:

- A **working implementation**
- Clear demonstration of all required OS concepts
- Modular and well-structured code
- Proper error handling and edge case consideration

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7. Deliverables

Students are required to submit:

7.1 Source Code (.zip / GitHub link)

- Complete and modular code
- Proper directory structure

7.2 Project Report

The report should include:

- Problem statement
- Explanation of how each required concept is implemented
- Screenshots/output
- Challenges faced and solutions

7.3 Demonstration and Viva

- Live demo of the project
- Explanation of implementation details
- **Viva will be conducted individually** to assess understanding of:
  - OS concepts used
  - Design decisions
  - Code implementation

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8. Evaluation Criteria

Projects will be evaluated based on:

- **Correct implementation of required concepts** (mandatory)
- Code quality and structure
- Innovation and complexity
- Demonstration and explanation
- Performance in **viva**

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9. Important Notes

- All required concepts **must be clearly visible in the implementation**.
- Plagiarism or copying will lead to **strict penalties**.
- Students are encouraged to build **practical, realistic, and innovative systems** (e.g., file systems, distributed apps, collaborative tools, etc.).
- **Originality in idea, design, or implementation will be positively considered during evaluation.**

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10. Suggested Project Ideas (Optional Inspiration)

- Distributed file sharing system
- Online collaborative editor
- Task scheduling system with concurrency handling
- Client-server based database system
- Online Banking System